Geophys. J. Int. (1999) **136,** 739–756

The palaeomagnetism of (Mesoproterozoic) Eriksfjord Group red beds, South Greenland: multiphase remagnetization during the Gardar and Grenville episodes

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Accepted 1998 October 15. Received 1998 October 9; in original form 1998 April 1

SUMMARY

The Eriksfjord Group comprises ~3000 m of lavas and sediments rapidly deposited in a rift which developed within an Andean-type batholith in juxtaposition to the southern margin of the Laurentian Shield in South Greenland at *ca.* 1300 Ma. The lavas have been shown to preserve a detailed record of the geomagnetic field at the time of eruption, incorporating normal, reversed and transitional directions. This study has examined the magnetic properties of the intervening red sediments. They are found to possess a diagenetic remanence imparted by mediating fluids at later times. The impact of diagenesis is stratigraphically controlled: the base of the rift infill has magnetizations partially resident in magnetite which are either unstable to thermal cleaning or record a single polarity 'B' magnetization ($D/I = 284/67^\circ$, 31 samples, $\alpha_{95} = 5.5^\circ$, palaeopole at 244.1°E, 47.5°N, $dp/dm = 7.5/9.1^\circ$). This corresponds in polarity, and closely in direction, to remanence observed in mid-Gardar gabbro giant dykes and dyke swarms emplaced along the axis of the rift system at *ca.* 1160 Ma; the causative diagenetic magnetite appears to have grown from hydrothermal systems motivated by this magmatism in a sealed reservoir setting within the lower part of the rift infill.

The Ilímaussaq alkaline igneous complex was emplaced into the southern extension of the rift at *ca.* 1130 Ma and possesses a dual polarity magnetization ($D/I = 327/81^\circ$, $\alpha_{95} = 6.4^\circ$, 10 sites). Eriksfjord lavas within the thermal aureole are overprinted to varying degrees by comparable magnetizations with steep inclinations. The mean pole position (283°E, 71°N, $dp/dm = 12/12^\circ$) lies near the apex of an apparent polar wander loop incorporating the Gardar Track (*ca.* 1300–1140 Ma) and the Keweenawan Track (*ca.* 1115–1050 Ma). Magnetizations in the Eriksfjord sedimentary succession have not been significantly reset by emplacement of the Ilímaussaq complex, but higher levels of the rift infill are dominated by an 'A' magnetization ($D/I = 305/34^\circ$, $\alpha_{95} = 4.3^\circ$, 57 samples, palaeopole at 202.1°E, 32.4°N, $dp/dm = 2.8/4.9^\circ$) resident in haematite. The pole position does not correspond with any part of the Gardar Track, but does correlate with the return Keweenawan Track at *ca.* 1090 Ma, close to the time of Grenville orogenesis along the bordering southeastern margin of the Laurentian Shield. This remanence is attributed to diagenesis during extensional tectonism linked to the collapse of the Grenville Orogen formerly sited 100–200 km to the south.

Key words: Gardar, Greenland, Grenville, Mesoproterozoic, palaeomagnetism, red sediments.

1 INTRODUCTION

Palaeomagnetic studies to unravel the dynamics of the continental crust during the long Proterozoic eon of Earth history (~2500–540 Ma) have focused on a number of rock suites, with each one having associated limitations. Metamorphic terranes, which were uplifted and magnetized following widespread orogenic/mobile episodes, can preserve an extended record of the palaeomagnetic field although this is difficult to date with precision. Igneous provinces may have well-constrained ages

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although they usually resulted from intense and short-lived magmatic events. Sedimentary-volcanic successions are sparsely preserved.

This is illustrated by the Laurentian (North America–Greenland) Shield (Fig. 1), where much of both the exposed and buried metamorphic basement was reworked by tectonothermal events between ~1900 and 1700 Ma in the 'Hudsonian' episodes. Smaller nucleii which escaped reworking comprise the Archaean cratons (>2500 Ma), some of which have a history extending back to ~3800 Ma. Later in Mesoproterozoic times, by which time they had been uplifted and cooled, the metamorphic terranes were subjected to brittle fracture in tensile stress regimes responsible for dyke emplacement, and to the formation of rift systems incorporating sedimentary-volcanic successions. Along the southeastern margin of the shield, the largest examples of the latter are the Keweenawan and Gardar rifts (Fig. 1). These events were precursors to the last Proterozoic mobile event to affect the Laurentian Shield, the Grenville Episode, at ~1100 Ma. The Grenville orogen was emplaced with a tectonic frontal zone sited 100–200 km to the southeast of present outcrop of the former rifts.

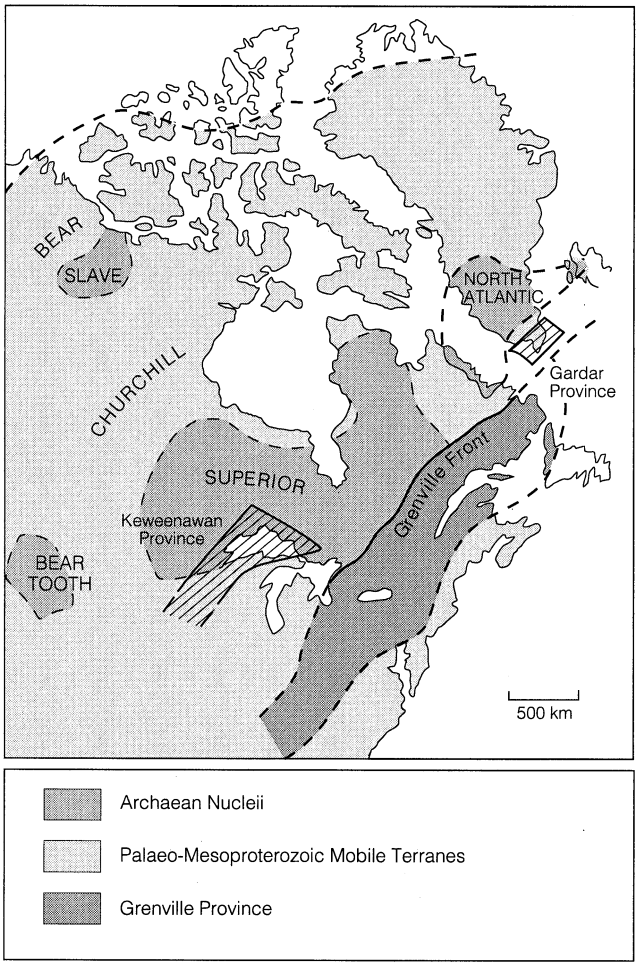


Figure 1. The Laurentian Shield of North America and Greenland showing the major tectonothermal divisions of the metamorphic basement and the locations of Mesoproterozoic rifts discussed in the text. Note that the NE–SW trends of the Gardar and Keweenawan rifts are approximately parallel to the inferred extensions of the Grenville Mobile Belt (*ca.* 1100 Ma).

The succession infilling the Gardar Rift in South Greenland comprises the Eriksfjord Group, approximately half of which is composed of lava successions. It has already been demonstrated that the lavas preserve a detailed record of the primary geomagnetic field at the time of formation (Piper 1977a; Thomas 1992; Thomas & Piper 1992, 1995). The present study has therefore focused on the sedimentary component of the Eriksfjord succession and aimed to compare the palaeomagnetic record preserved in the two contrasting facies that make up this Group.

2 GEOLOGICAL FRAMEWORK

In the Gardar Province of South Greenland, the rift infill is associated with intraplate alkaline igneous complexes, major gabbroic bodies and dolerite dyke swarms; these were more widely emplaced between 1320 and 1120 Ma across a belt at least 200 km long and 70 km wide (Emeleus & Upton 1974). The Gardar province is a component of a long-lived tectonomagmatic zone extending southwestwards via the Great Abitibi Dyke (1140 Ma) and associated swarms, to the Keweenawan Province (Macdonald & Upton 1993; and Fig. 1); the latter province is sited at the northwestern extremity of the Mid-Continental Rift Sytem with a subsurface extent traversing some 2000 km of the Laurentian Shield.

Rifting and magmatic activity in South Greenland during the Gardar interval were concentrated close to the boundary between the Archaean craton (>3500 Ma) and the Ketilidian mobile belt, an orogen skirting the southern tip of Greenland. This belt consolidated between 1800 and 1600 Ma, some 300 Ma before the commencement of Gardar activity. The rift succession is preserved within a 15–20 km wide palaeorift now defined by major NNE-trending fjords believed to lie along the boundary faults (Fig. 2). This (Tugtutôq–Ilímaussaq) lineament is coincident with a strong coaxial gravity anomaly (Blundell 1978); it was the focus of much dyking later in Gardar times and appears to have been the site of lithospheric thinning with high-level rifting complemented by emplacement of a mafic magma body at, or near to, the base of the crust (Macdonald & Upton 1993). Sediment deposition, comprising about half of the ~3000 m thick rift infill, was interrupted by effusion of lavas which progressively dominate the succession upwards. These are probably the oldest record of Gardar magmatism and comprise three formations (Mussartût, Ulukasik and Ilímaussaq). The majority (~70 per cent) are of basaltic and hawaiitic composition (Upton & Emeleus 1987), but trachyte lavas (~23 per cent) become important at higher levels. The three lava formations show a general increase in differentiation with time, suggesting derivation from a single magma chamber, or closely related nest of chambers, that evolved compositionally; such volcanic systems rarely have lifespans in excess of ~5 Myr.

Thus infilling of the Gardar rift probably occurred very rapidly in geological terms: the sediments are predominantly coarse clastics, mostly arenitic sandstones with subordinate conglomerates, and no unconformities are known (Poulson 1964). Well-preserved pahoehoe lava surfaces also point to high eruption rates, and there is a general absence of erosion and palaeosol development between flows (Upton 1974). The middle (Ulukasik) lava succession was mostly emplaced within a single polarity transition of the ancient geomagnetic field

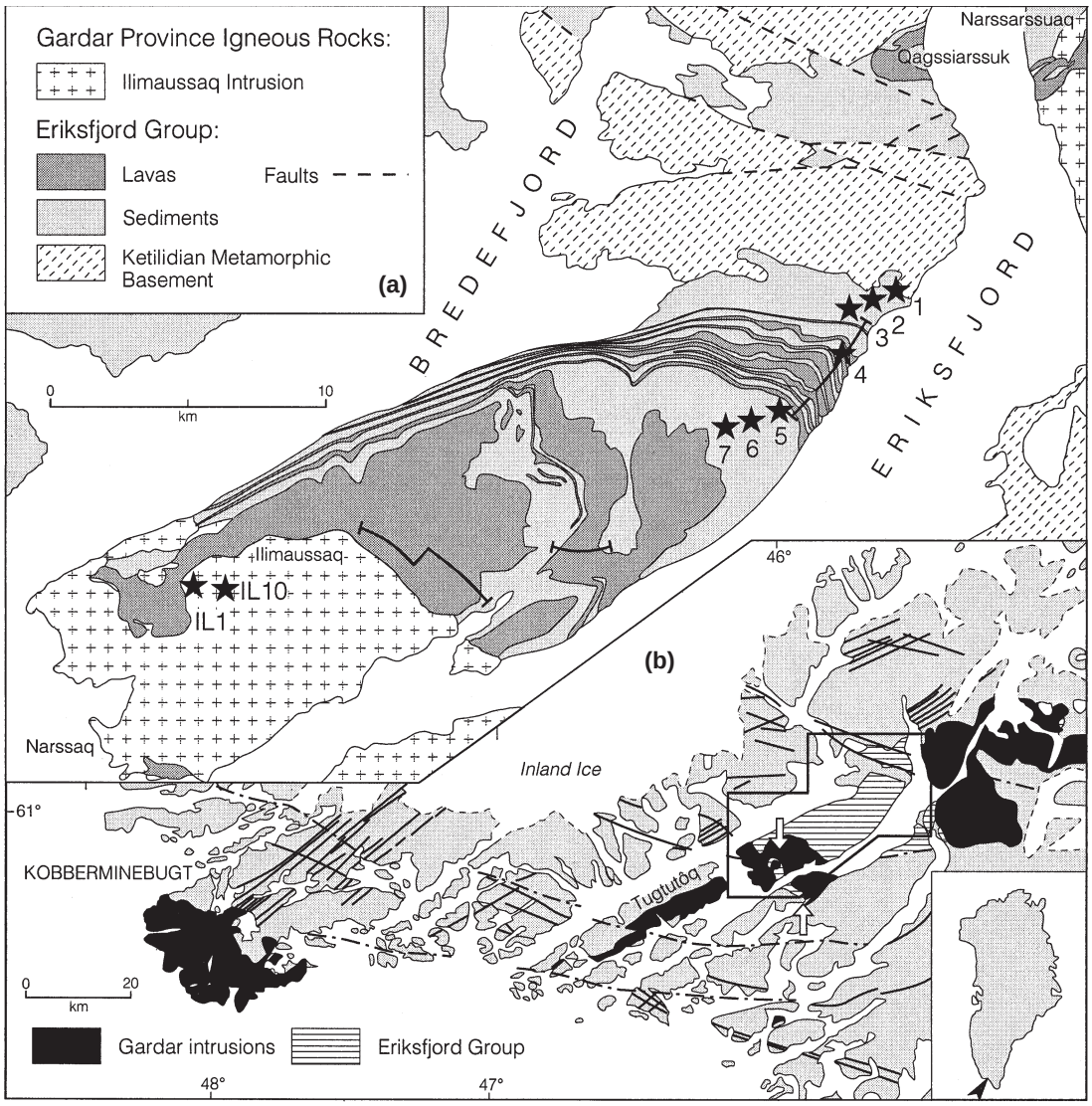


Figure 2. (a) Geological map based on the 1:100 000 map of Greenland (Berthelsen & Henriksen 1975) showing a sector of South Greenland incorporating sedimentary and volcanic infill of the Gardar Rift. The stars show locations of palaeomagnetic sampling sites in the red beds and Ilímaussaq intrusion; the continuous lines cutting the lavas outcrop are palaeomagnetic traverses through the volcanic successions yielding a polarity stratigraphy described by Thomas & Piper (1992, 1995). (b) Outline geological map of the Gardar province after Upton & Emeleus (1987) showing the location of the rift in the context of the wider igneous outcrop of central complexes and dyke swarms. The arrows show sampled locations in the Ilímaussaq complex discussed in the text. Note that this intrusion (1130 Ma) post-dates the complex of giant dykes and minor dyke swarms (1160–1154 Ma) running down the axis of the island of Tugtutôq to the SW.

(Thomas & Piper 1995) and must therefore record an eruptive event lasting no more than a few thousands of years.

The six formations comprising the Eriksfjord Group are listed below from top to bottom, with representative stratigraphic thicknesses after Stewart (1964):

Ilímaussaq Volcanic Formation 1030 m

*Nunasarnaq Formation 350 m

Ulukasik Volcanic Formation 150 m

*Naujurssuit Formation (sites 7–8) 420 m

Mussartût Volcanic Formation (site 6) 745 m

*Mâjût Sandstone Formation (sites 1–5) 390 m

The sedimentary formations (formerly referred to as 'members' in the descriptive literature, see Poulson 1964 and Allaart 1973) are indicated with an asterix and form the focus of this study; in addition, lava emplacement in the Mussartût Formation was interupted by deposition of sandstones and conglomerates, with the former yielding site 6. The sediments are entirely of continental origin and deposited mainly in a fluviatile environment; subordinate lacustrine and aeolian facies are also reported. The sandstones are composed predominantly of quartz and feldspar. The red pigment, which is widely developed up to the level of the Ulukasik Formation, is bleached where overlain by lava flows and at the margins of ~1150 Ma dykes; accessory haematite was therefore present (but not necessarily fixed) during Gardar times. Although

they preserve a primary magnetic remanence, the lavas of the Eriksfjord Group have all been subjected to low-grade alteration with widespread development of epidote, implying thermal metamorphism to temperatures of 300–350 °C.

The age of the Eriksfjord Group is poorly constrained. It rests unconformably on Ketilidian granites dated at 1810–1780 Ma with a final cooling age of ~1600 Ma (van Breemen *et al.* 1974), and northern outliers of the Group are cut by the Motzfeldt [1310 ± 31 Ma (Rb–Sr), 1350 ± 10 Ma (Sm–Nd)] and North Qôroq [1299 ± 61 Ma (Rb–Sr)] plutons (Blaxland *et al.* 1978; Cassi *et al.* 1993). It is possible that the higher lavas are related to emplacement of these complexes (Upton & Emeleus 1987); if this is the case, the entire succession may be ~1350–1310 Ma in age and some 150–300 Ma older than the Mid-Continental Rift System within the Laurentian Shield (Fig. 1). There are two Sm–Nd determinations of 1200 ± 30 and 1170 ± 30 Ma on lavas of the middle (Ulukasik) formation of the Eriksfjord Group (Cassi *et al.* 1993), but these appear to be inconsistent with the geological and palaeomagnetic constraints and probably record partial resetting during mid-Gardar igneous activity (Section 7.2).

The Eriksfjord Group is extensively cut by NNE-trending giant and minor dykes, and the SW extension is obscured by emplacement of plutons in the Narssaq region; all of this magmatism appears to have been concentrated within the interval 1160–1130 Ma (Blaxland *et al.* 1976; Cassi *et al.* 1993). The last major igneous event to influence the present outcrop (see Fig. 2) was the emplacement of the Ilímaussaq alkaline complex dated 1143 ± 21 Ma (Rb–Sr) and 1130 ± 50 (Sm–Nd).

3 FIELD AND LABORATORY METHODS

Samples were cored in the field using a portable drill and orientated by Sun and magnetic compasses along a sampled section through the Eriksfjord Group shown in Fig. 2. Five sites are in sandstones of the basal Mâjût Formation, one site (6) is from a sandstone intercalated with lavas of the Mussartût Formation, and two sites are from red sandstones of the Naujarssuit Formation. The highest sedimentary (Nunasarnaq) formation is composed of white arenites of aeolian origin; these are now thoroughly cemented to quartzites and were not sampled for this study.

Field cores were prepared into cylinders of length 2.2 cm and their natural remanent magnetizations (NRMs) measured by a SQUID magnetometer. Progressive thermal demagnetization was then conducted in steps of 50 °C up to 500 °C and subsequently in steps of 10, 15 or 20 °C to 720 °C, or until either the remanence ceased to be measurable or directional behaviour was no longer systematic. Magnetic susceptibility was measured regularly during the treatment to monitor chemical changes induced by heating. All measurements and treatments were conducted within a nullified field space provided by a large set of Rubens coils to minimize secondary remanence acquisition. Following completion of the demagnetization, components comprising the NRM were identified by studying orthogonal projections of the remanence vector. Individual magnetizations were then isolated and their directions calculated by principal component analysis.

4 PALAEOMAGNETISM OF THE ERIKSFJORD GROUP RED BEDS

Palaeomagnetic studies of the lava successions within the Eriksfjord Group have demonstrated the presence of multiple reversals of the ancient geomagnetic field. A coherent magnetic stratigraphy of normal and reversed zones, together with the recognition of localized overprinting in the aureoles of ~1150 Ma plutons and dykes (Piper 1977a; Thomas 1992; Thomas & Piper 1992, 1995), demonstrates that the lavas possess a primary thermomagnetic remanence (TRM). The ensuing stable cratonic setting and the remoteness from later tectono-thermal episodes have no doubt facilitated preservation of this magnetostratigraphy. The dual polarity remanence preserved in the higher part of the lava succession is, however, not coaxial, a phenomenon observed elsewhere in rocks of late Mesoproterozoic age (Nevanlinna & Pesonen 1983).

The lavas of the Mussartût Formation have normal polarity exclusively and a mean remanence direction of $D/I = 270/22^\circ$ (Thomas & Piper 1992). Note that the 'normal' assignment is used here to refer to the positive inclination in the northern hemisphere and does not imply that this was the specific polarity of the Mesoproterozoic field. By the time that the higher Ilímaussaq Formation lavas were erupted, this direction had moved clockwise to a steeper inclination ($D/I = 296/41^\circ$) whilst the lavas of opposite polarity yield a shallower mean of $D/I = 97/-15^\circ$.

We observe two key differences between remanence in the lavas and sediments. First, the sedimentary record is not of uniform stability: many samples from the lower sediments in the Mâjût Formation are poorly stable or show no coherent remanence record. Second, the typical most stable remanence declination in the sediments is exclusively of normal polarity and rotated further clockwise compared with declinations recorded in the lavas.

These properties prove to be stratigraphically related. Fig. 3 illustrates examples from sites 1–4 in the Mâjût Formation. The commonest feature of the remanence structure is a distributed blocking temperature component ('B') of westerly declination and intermediate to steep inclination. This component is removed by the Curie point of magnetite in some samples (1-3 and 1-35 in Fig. 3), whilst in others the demagnetization behaviour becomes erratic at ~550° and higher blocking temperature components are not recoverable. The 'B' component does not converge to the origin of the orthogonal plots in all samples (see, for example, 1-24, 1-27 and 4-2 in Fig. 3), although magnetomineralogical changes above 590 °C preclude recovery of residual high blocking temperature components. At sites 2 and 3 chemical changes occurred at lower temperatures, and only a few samples yielded component definition. A subsidiary feature of site 4 is the presence of distributed blocking temperature components with S+ direction (samples 4-7 and 4-8 in Fig. 3) as well as the WNW+ components (sample 4-2); the S+('C') component is also present as a lower blocking temperature component in some other samples (1-27 in Fig. 3). Magnetomineralogical changes are observed in samples containing the 'C' component as the magnetite Curie point is approached (example 4-7 in Fig. 3).

The alteration evident from erratic directional and intensity behaviour in the Mâjût Formation is mirrored in susceptibility changes with temperature; these take the form of a large decrease or increase above ~400 °C (Fig. 4). samples across a sedimentary slump structure at site 1 show only a homogenous 'B' magnetization and identify this remanence as post-dating soft sediment deformation. A comparison was also made

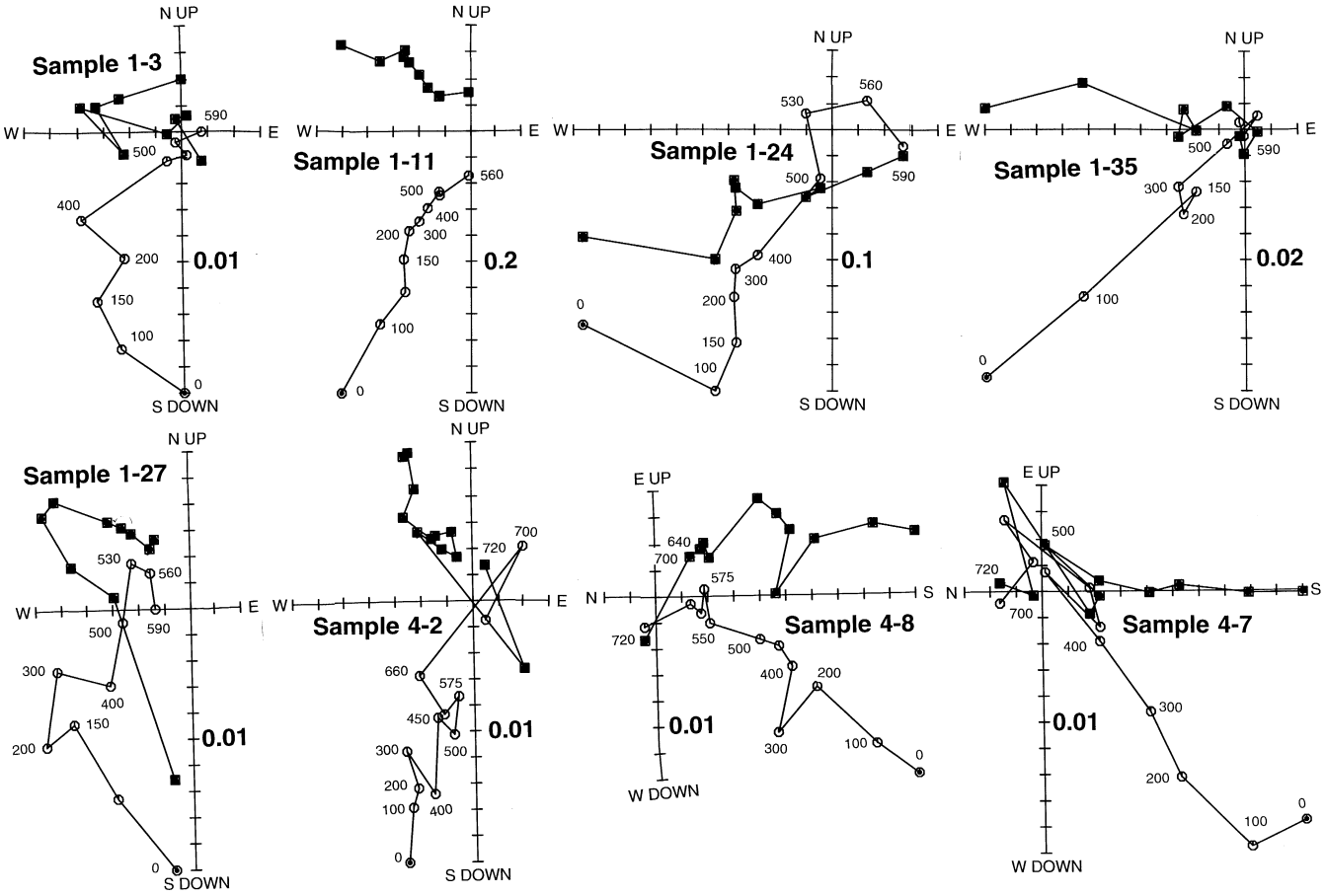


Figure 3. Orthogonal projections illustrating examples of progressive thermal demagnetization of red bed samples from the Mâjût Formation. Projections onto the horizontal plane are shown as closed squares, and projections onto the vertical plane are shown as open circles. Intensities of magnetization are $\times 10^{-5}$ A m^2 kg^{-1}.

between red and bleached samples at this site. Although no contrasting component structures were observed, bleached samples are between one and two orders of magnitude more weakly magnetized than the red samples; hydrothermal leaching, which is also observed here along the contacts of $\sim$1150 Ma dykes, has therefore either removed iron minerals or converted them to paramagnetic phases.

With a few exceptions, samples from red sediments in the higher Mussartût and Naujurssuit Formations are much less susceptible to change during the thermal demagnetization (*cf.* sample 5-7 in Fig. 5). This is demonstrated by little or no change in susceptibility with temperature (samples 6 and 7 in Fig. 4). The dominant feature of the stratigraphically highest sites 7 and 8 is a NW-directed component of shallow to intermediate positive inclination ('A') which is often discrete and is only fully unblocked near to the Curie point of haematite (Fig. 5). This component is also represented at sites 5 and 6, but is present solely in the magnetite spectrum (sample 5-1 in Fig. 5) or removed prior to the recovery of a higher blocking temperature component (6-12). Other samples at these two sites have the WSW+ components, in this case only subtracted above the Curie point of magnetite (6-1 in Fig. 5), and occasional examples of the S+'C' magnetizations (samples 6-4 and 6-12 in Fig. 5).

Component directions assigned to the 'A', 'B', and 'C' populations are shown in Fig. 6, and site and group mean palaeomagnetic results are summarized in Table 1. The 'A' population is confined to the Mussartût and Naujurssuit formations; the 'B' population is found predominantly in the Mâjût Formation with rare examples in the higher formations. The 'C' component is observed in isolated cores throughout. The Eriksfjord succession dips at low angles averaging 10° in a SSW direction in the sample area. Some of this tilt was probably acquired by downsag during initial infill, although part is related to faulting and some may post-date the Ilímaussaq complex. Since, as we argue below, remanence in the sediments is a diagenetic phenomenon acquired in late to post-Gardar times, it is probable that it post-dates most or all of the tilting; accordingly, we treat the mean directions *in situ* although this assumption has no substantial effect on their interpretation.

5 ROCK MAGNETISM OF THE RED BEDS

The complex magnetization histories that characterize most red beds are a consequence of prolonged lithification and diagenesis. Various phases of haematite are the usual products of these processes in oxidizing environments and may include detrital specularite, pigmentary films and authigenic grains and overgrowths. Because of their uniformly high coercivities and common high Curie point close to 680 °C, these phases

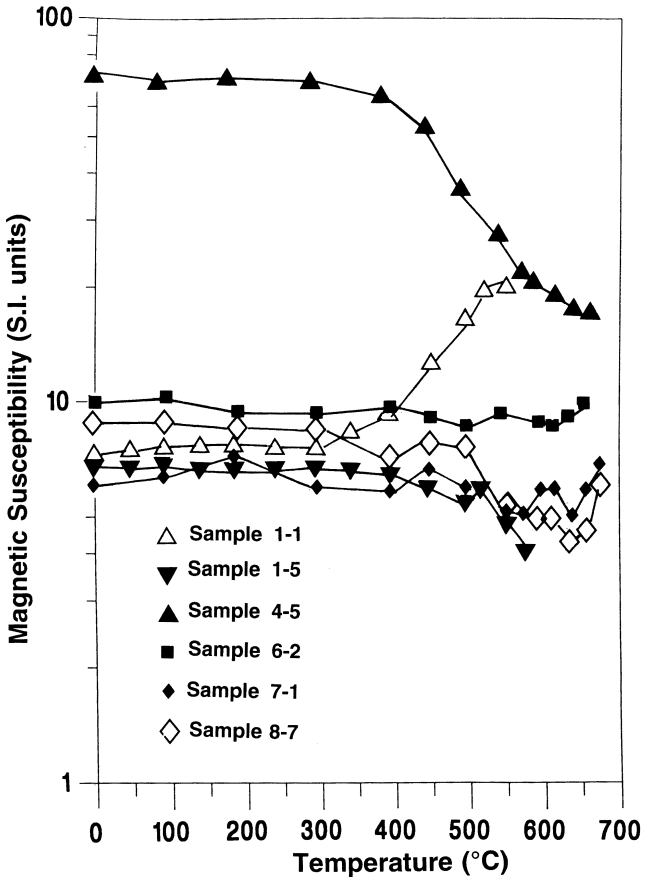


Figure 4. Susceptibility changes in the red sediments as a function of temperature during the thermal demagnetization.

Table 1. Site and group mean palaeomagnetic results from red beds of the Eriksfjord Group, South Greenland.

Site No.	Component	N/n	D	I	R	k	α_{95}
1	B	45/24	274.6	67.4	23.20	28.8	5.6
1–3	B	24/7	309.5	60.3	6.69	19.6	14.0
3	C	15/8	168.4	38.0	7.83	42.0	8.6
4	C	15/4	193.4	31.7	3.86	21.7	20.2
5	A	8/6	306.8	47.7	5.81	26.6	13.2
6	A	15/6	303.1	32.6	5.87	37.4	11.1
7	A	25/20	309.3	19.6	19.61	49.1	4.7
8	A	14/14	305.6	26.6	13.20	16.3	10.2
Group Mean Results							
		D	I	N	R	k	α_{95}
A magnetizations		304.6	33.5	57	54.2	20.1	4.3
(palaeomagnetic pole: 202.1°E, 32.4°, $dp/dm = 2.8/4.9°$)							
B magnetizations		283.8	66.6	31	29.68	22.7	5.5
(palaeomagnetic pole: 244.1°E, 47.5°N, $dp/dm = 7.5/9.1°$)							
C magnetizations		178.9	38.2	17	16.22	20.6	8.1
(palaeomagnetic pole: 315.4°E, 7.5°S, $dp/dm = 5.7/9.5°$)							

D and I are the mean declination and inclination, respectively, derived from n components out of a total population of N cores. The resultant vector is R, and k is the Fisher precision parameter $[=(n-1)/(n-R)]$. α_{95} is the radius in degrees of the 95 per cent cone of confidence about the mean direction. dp and dm are the radii of the oval of 95 per cent confidence about the palaeomagnetic pole position in the colatitude direction and at right angles to it, respectively.

are not well discriminated by most rock magnetic techniques. Isothermal remanent magnetization (IRM) studies are amongst the most sensitive for detecting trace concentrations of magnetic minerals (Robertson & France 1994) and have been applied in this study to highlight types and relative amounts of magnetic minerals.

Selected samples were magnetized using a pulse magnetizer, and applied fields were increased in approximately logarithmic steps (2, 5, 10, 15, 20, 25, 30, 40, 50, 80, 100, 120, 120, 160, 200, 250, 300 mT …) until an upper limit of 2500 mT was reached. Although the peak field in the pulse demagnetizer is unable to fully saturate minerals with high coercivities such as haematite, it is adequate to identify the essential behaviour of this mineral. Subsequently, samples were incrementally magnetized by the same process in the opposing direction until the IRMs were reduced to zero. At this point the applied field value defines the coercivity of the sample remanence (H_{cr}).

IRM acquisition curves (Fig. 7) are similar but indicate significantly higher coercivities in samples from the Mussartût and Naujurssuit formations compared to results from the Mâjût Formation. IRM acquisition is best defined by a cumulative log Gaussian (CLG) function: when multiple phases are present, the acquisition curve can be described by a combination of the individual CLG functions (Eyre 1994). The Eriksfjord red beds all yield curves showing a combination of at least two cumulative functions; the gradient of the acquisition curve can then be plotted against the log of the applied field using the method proposed by Eyre (1994). Three parameters describe each GLC function and are listed in Table 2. The SIRM is the height of the log Gaussian curve, $B_{1/2}$ is the applied field at which 50 per cent of the SIRM is acquired, and DP is a statistic describing the dispersion of apparent coercivities of the magnetizing phase. The parameters $B_{1/2}$ and DP are intrinsic properties related to the type of mineral and are independent of concentration (Robertson & France 1994), whereas SIRM is extrinsic and volume-dependent; it therefore depends on the dilution factor of the magnetizing phase within the sample.

Values of $B_{1/2}$ and DP are similar for the two phases in each of the tested samples. The higher values may confidently be attributed to the haematite; the lower values cannot be uniquely assigned to a particular ferromagnetic mineral but, in the context of the components unblocked near 580 °C, are probably due to low-Ti titanomagnetite. Both the absolute (see SIRM values in Table 2) and relative (Fig. 8) amounts of haematite are much larger in the stratigraphically higher formations. The greater importance of haematite is further indicated by the higher unblocking temperature spectra at these higher-level sites (see J_{400} in Table 2).

Relative contributions of high- and low-coercivity components to the remanence can be evaluated by thermal demagnetization of an orthogonal three-component IRM (Lowrie 1990). IRMs of 2.5, 0.4 and 0.1 T are applied in that order along mutually orthogonal directions and are referred to as hard, medium and soft components, respectively. Fig. 9 shows an example from each of the three formations and illustrates the increasing importance of the hard component in the higher formations. In each case the hard component is unblocked between 600 and 650 °C, suggesting that the haematite is titaniferous.

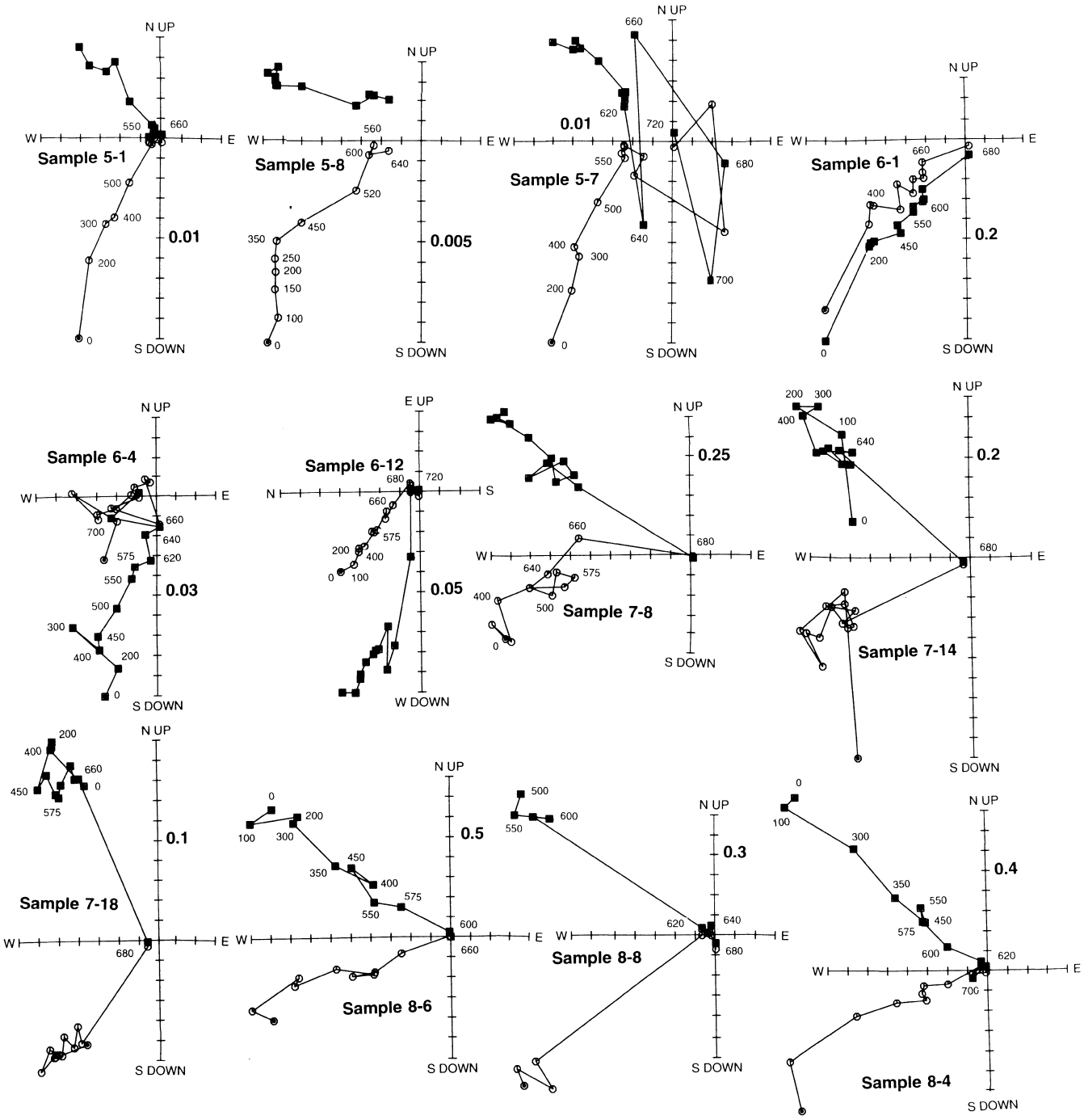


Figure 5. Orthogonal projections illustrating examples of progressive thermal demagnetization of red sediments from the Mussartût and Naujurssuit formations. The symbols are as for Fig. 3.

These rocks are terrestrial sediments deposited in a weakly oxidizing Mesoproterozoic environment. Since they are composed mainly of quartz and feldspar derived from the granitic basement bordering the palaeorift, it seems likely that the haematite originated from the breakdown of less stable paramagnetic silicates such as biotite (see for example, Turner & Archer 1977). The magnetite could have been derived directly from an igneous or metamorphic source, in this case predominantly granitic. The remanence resident in both the haematite and magnetite was, however, fixed at some later stage: whilst the polarity of the 'A' and 'B' remanences is comparable to the lower (Mussartût Formation) lavas with which these sediments are interleaved (only three flows of reversed polarity have been identified, see Thomas & Piper 1992), the mean directions of magnetization are significantly different. A mean remanence direction of $D/I = 270/22^{\circ}$ in the lavas contrasts with the declination of the 'A' magnetization $D/I = 305/34^{\circ}$, and with the much steeper inclination ($I = +67^{\circ}$) of the 'B' magnetization which dominates the lower red sediments interleaved with these lavas. The difference is more marked when comparison

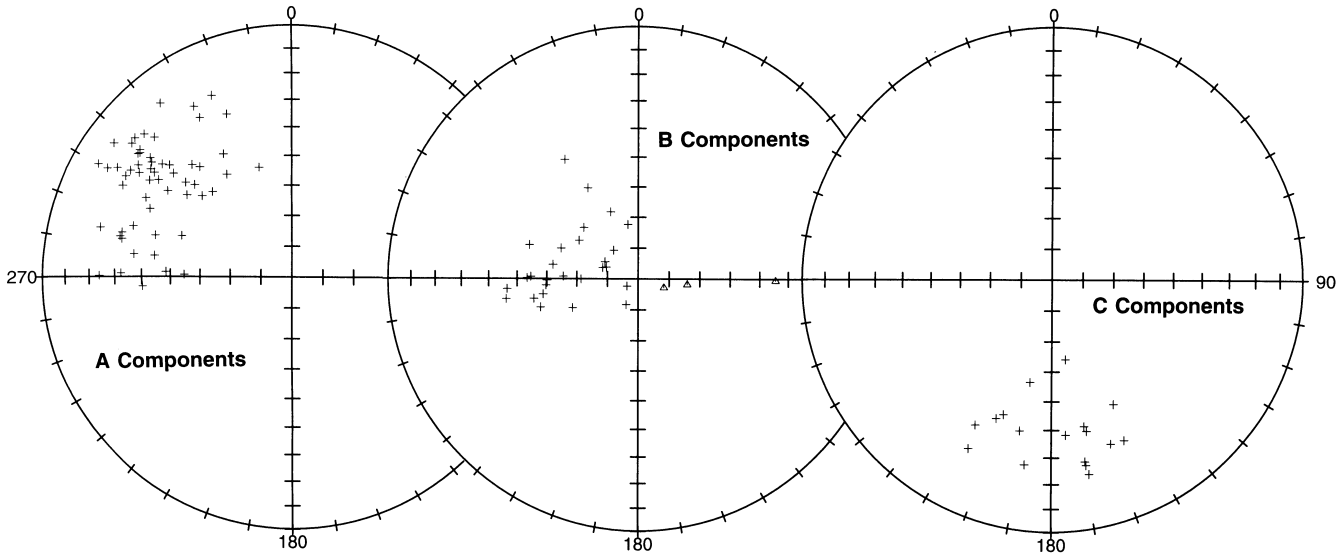


Figure 6. Directions of components assigned to the 'A' (Mussartût and Naujurssuit formations), 'B' and 'C' populations. Crosses are lower-hemisphere plots and open triangles are upper-hemisphere plots; equal-area projections.

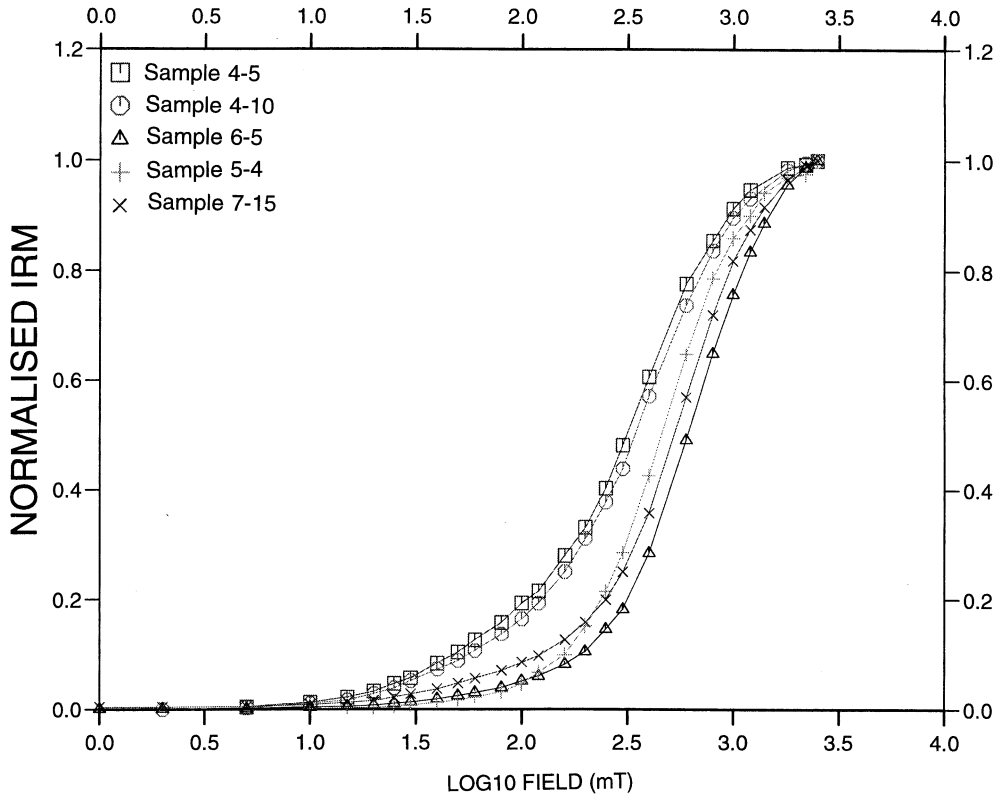


Figure 7. IRM acquisition curves from samples of Eriksfjord Group red sediments.

is made with lavas at the base of the Eriksfjord succession in outliers to the north at Narssassuaq ($D/I = 290/12°$, Piper 1992). The mildly reducing and alkaline conditions required to precipitate magnetite during diagenesis are most likely to have prevailed during subsequent burial. Since sedimentation occurred near the commencement of the Gardar tectono-magmatic episode, subsequent igneous events in the proximity of the rift will have driven hydrothermal circulation through these formations and may have played a key role in mobilizing iron and depositing authigenic magnetite or haematite. In addition, the lava formations have been subjected to low-grade metamorphism during burial to temperatures of 300–350 °C. Since mineralogical effects of this metamorphism in the lavas are mainly confined to precipitation of paramagnetic silicates such as epidote, it has had no apparent impact on their remanence. The consequences of a comparable metamorphism within the porous and weakly lithified sediments are, however, likely to have been very different (Section 7).

Table 2. Results of unidirectional IRM analysis.

Sample No.	First Mineral Phase SIRM ($A\,m^2\,kg^{-1}$)	Log $B_{1/2}$ (mT)	DP log(mT)	Second Mineral Phase SIRM ($A\,m^2\,kg^{-1}$)	Log $B_{1/2}$ (mT)	DP log(mT)	H_c	J_{400}*
4–5	0.0000	1.80	0.29	0.0001	2.60	0.25	320	0.367
4–10	0.0000	1.60	0.31	0.0002	2.60	0.35	351	0.514
5–4	0.0001	1.90	0.45	0.0005	2.79	0.29	597	0.783
6–45	0.0002	1.90	0.45	0.0015	2.83	0.30	644	0.767
7–15	0.0002	1.85	0.40	0.0038	2.66	0.29	517	0.875

* J_{400} is the fraction of the natural remanent magnetization ($\times 10^{-5}\,A\,m^2\,kg^{-1}$) remaining after thermal demagnetization to 400 °C.

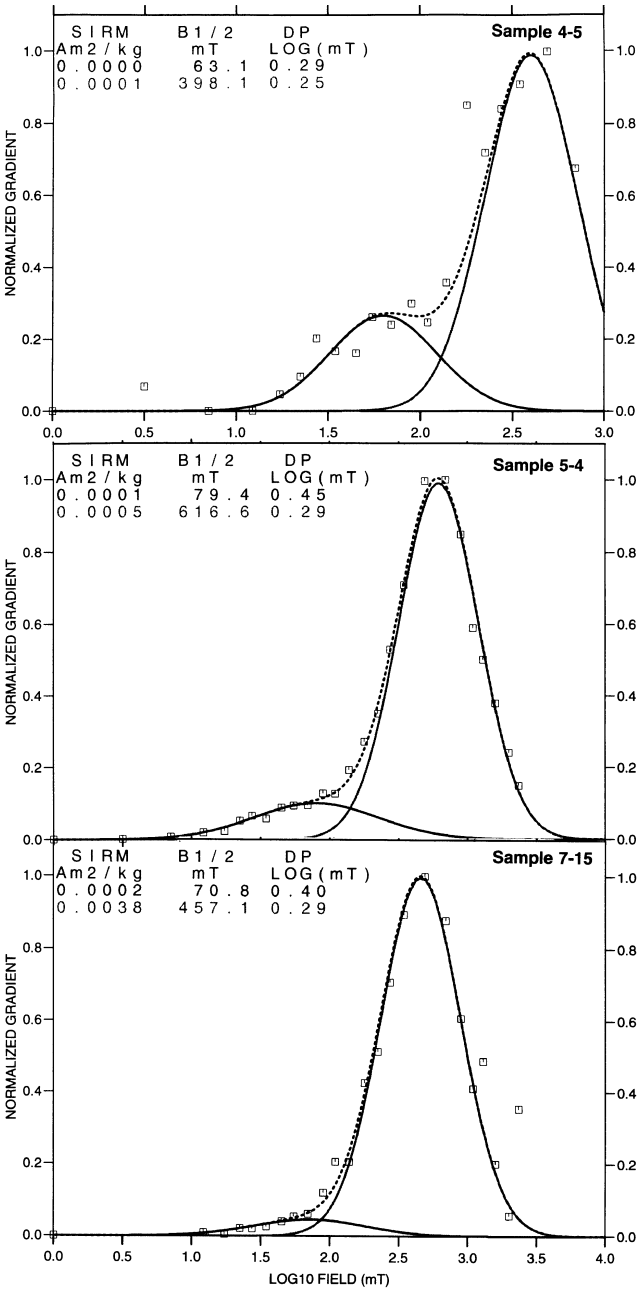


Figure 8. Gaussian functions fitted to the normalized gradient of IRM in samples from each of the three sampled stratigraphic levels within the Eriksfjord Group.

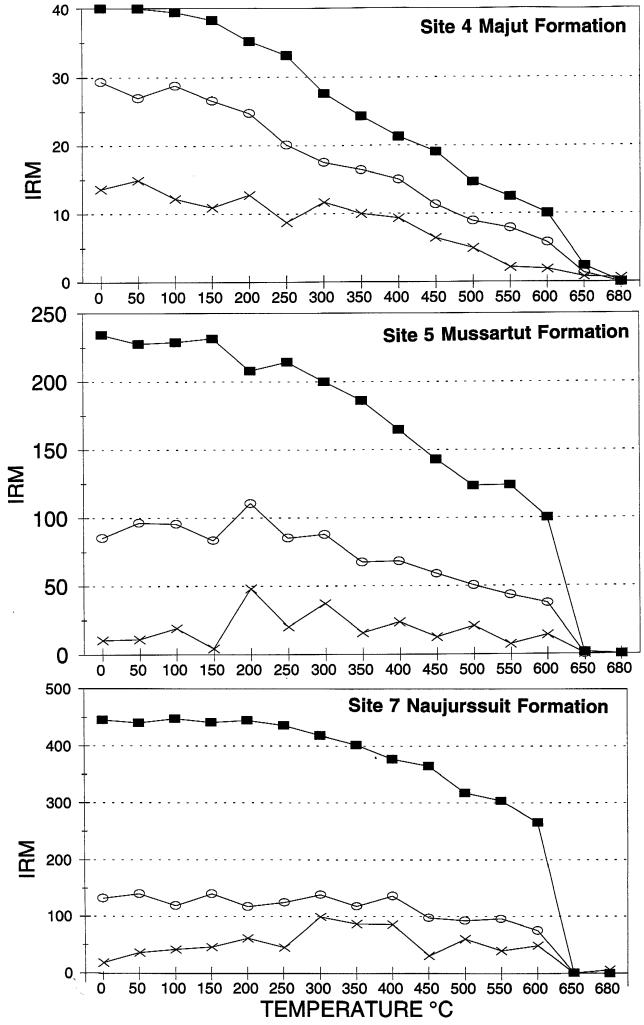


Figure 9. Examples of thermal demagnetization of three-component IRMs from the three sampled stratigraphic levels within the Eriksfjord Group. The squares are hard (high coercivity) components, the ovals are medium and the crosses are soft components.

6 PALAEOMAGNETISM OF THE ILÍMAUSSAQ ALKALINE INTRUSION

This intrusion straddles the Gardar rift (Fig. 1) and is the largest igneous complex cutting the Eriksfjord Group. The earliest intrusive phase is an augite syenite forming a marginal envelope to the pluton. An interior undersaturated alkaline

(agpaitic) facies shows high degrees of fractionation (Sorensen *et al.* 1969; Larsen & Sorensen 1987): nepheline appears in syenite away from the contact, and in the interior of the intrusion rhythmically layered kakortokites (eudialyte-bearing nepheline syenites) occur in some 29 major layers (Bohse *et al.* 1971). Since xenoliths of syenite are present in the layered agpaitic rocks, the marginal syenite carapace is inferred to have solidified before the agpaitic rocks differentiated at depth in an enclosed parent syenite magma reservoir (Hamilton 1964; Ferguson 1964). Age determinations of 1143 ± 28 Ma by Rb–Sr (Blaxland *et al.* 1976) and 1130 ± 50 Ma by Sm–Nd (Cassi *et al.* 1993) imply that emplacement of this intrusion is one of the last igneous episodes within the Gardar Province.

Palaeomagnetic study of the southern part of the Ilímaussaq pluton at Tunugdliarfik (see Fig. 2b) has recognized stability to alternating field (a.f.) demagnetization in the marginal syenites and bordering kakortakites (Piper 1976). Declining intensities of magnetization, lower coercivities and poor stabilities were observed into the more fractionated facies in the interior of the body. A.f. demagnetization and end-point analysis of cores from marginal syenites yielded a mean direction of $D/I = 253/70^\circ$ (five sites, $\alpha_{95} = 11$). We have since conducted further sampling of this intrusion at 10 sites in the northern part of the body (sites IL1–10 in Fig. 2a). The agpaitic facies, which were only poorly stable to a.f treatment (Piper 1976), proved to be unstable to thermal demagnetization (Thomas 1992), probably because these rocks contain accessory pyrite and pyrrhotite (Ferguson 1964) which oxidizes during this treatment. The marginal syenites, however, are mostly stable to thermal demagnetization and define ChRMs unblocked by the Curie point of the magnetite (Fig. 10a).

Unfortunately, these northerly steep positive directions are indistinguishable from the present geomagnetic field direction at this location (Fig. 10b). Most vector plots show slight steepening or shallowing of the vector trajectory following demagnetization to 100 or 200 °C (see samples IL2-1, 3-7 and 5-8 in Fig. 10a), suggesting that a present field component has been subtracted. It seems unlikely that the higher blocking temperature component, sometimes stable up to 620 °C, could have been totally acquired in the Recent field, and we suspect that the magnetic field during primary cooling at ~1130 Ma was, fortuitously, close to the present field direction.

To test this proposition the demagnetization behaviours of Ilímaussaq Formation lavas of the Eriksfjord Group within the thermal aureole of the pluton were also examined (see sample profile in Fig. 2). This lava succession exhibits

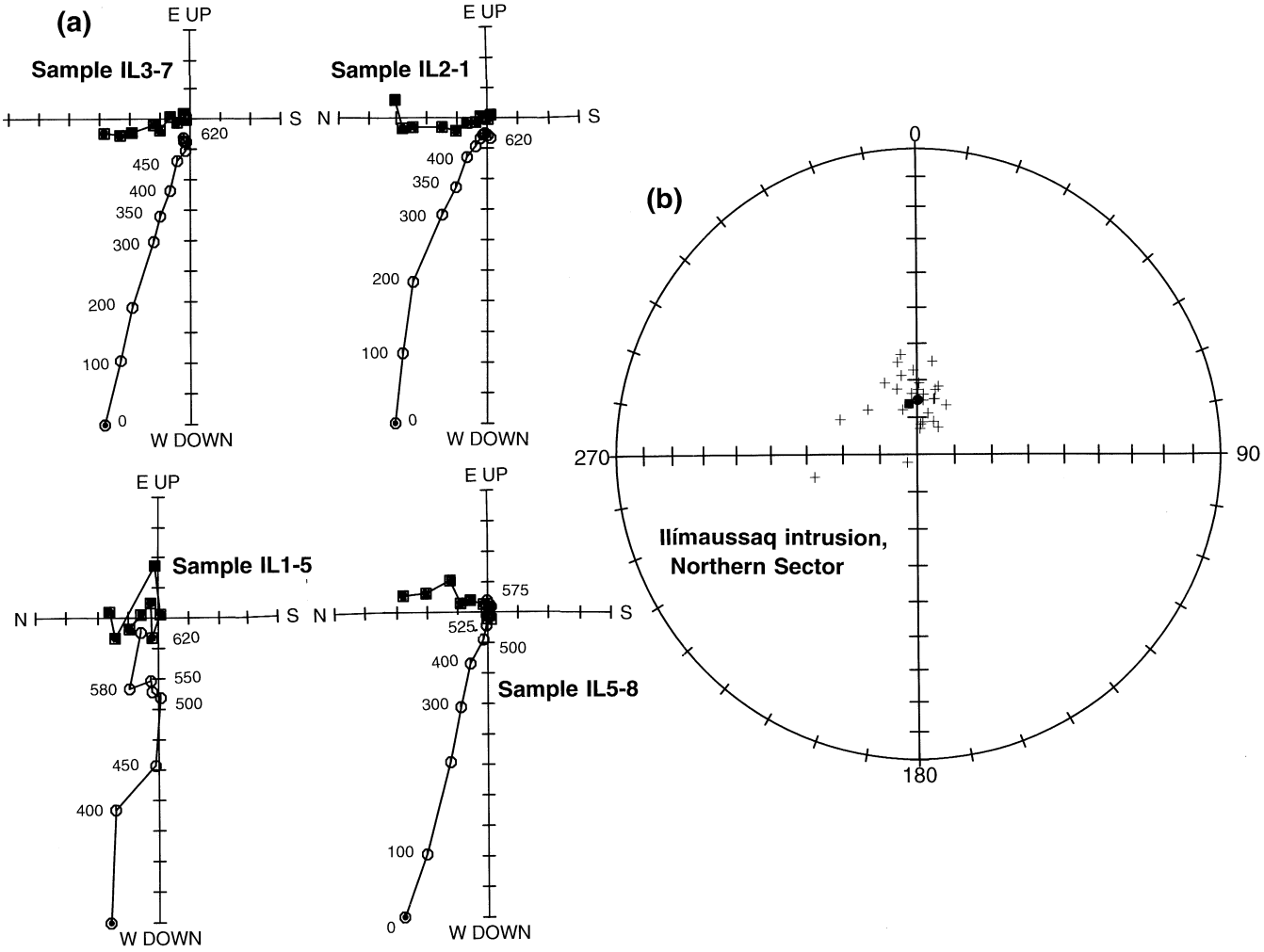


Figure 10. (a) Orthogonal projections illustrating examples of thermal demagnetization of syenite samples at sites IL1–5 from the northern part of the Ilímaussaq intrusion (see Fig. 2a); symbols are as for Fig. 3. (b) Sample ChRM directions from syenites at sites IL1–5. The closed square is the group mean direction and the circle is the present-day field direction.

four normal and three reversed polarity zones (Thomas & Piper 1992), and easterly declination and shallow to intermediate inclinations (see samples 14-1 and 44-3 in Fig. 11). The remanence is typically a discrete single component and has high unblocking temperatures, often above the Curie point of magnetite. The first signature of overprinting by a steep positive magnetization is observed at 1100 m from the margin of the pluton (Thomas 1992). Within 500 m of the contact more than half the lava samples are overprinted, although even within 300 m of the margin it is still possible to recover the primary polarity of the flows because some samples continue to preserve the high unblocking temperature deuteric remanence.

The character of overprinting is therefore sporadic; in the sampled sector beyond 250 m from the contact it is more likely to have been promoted by hydrothermal flow than by thermal unblocking. Where both primary and overprinted magnetizations are preserved, the primary magnetization is usually preserved in the lower blocking temperature component, whilst the overprint is unblocked mostly or entirely above the Curie point of the magnetite (samples 43-5 and 46-4 in Fig. 11), indicating that haematite deposition from hydrothermal circulation during reheating of the lava pile is responsible. The total overprints are steep positive components residing in both magnetite and haematite. Their declinations are variable and range from west to north to northeast (see samples 59-3, 59-1, 20-4 and 60-7 in Fig. 11). The mean of 38 overprinted ChRMs from lavas in the aureole is $D/I = 313/78^\circ$ ($\alpha_{95} = 5.1^\circ$, Table 3).

Thus steep positive inclinations are a feature of both the marginal syenites at sites IL1–5 and overprints in the aureole. The declinations of the overprints are more variable and include westerly values comparable with westerly declination resolved from the southern outcrop of the pluton (Piper 1976); in contrast, steep northerly positive directions were recognized in the interior facies of the pluton in this region, although stability was poor and the possibility of viscous remagnetization in the present geomagnetic field could not be excluded. These

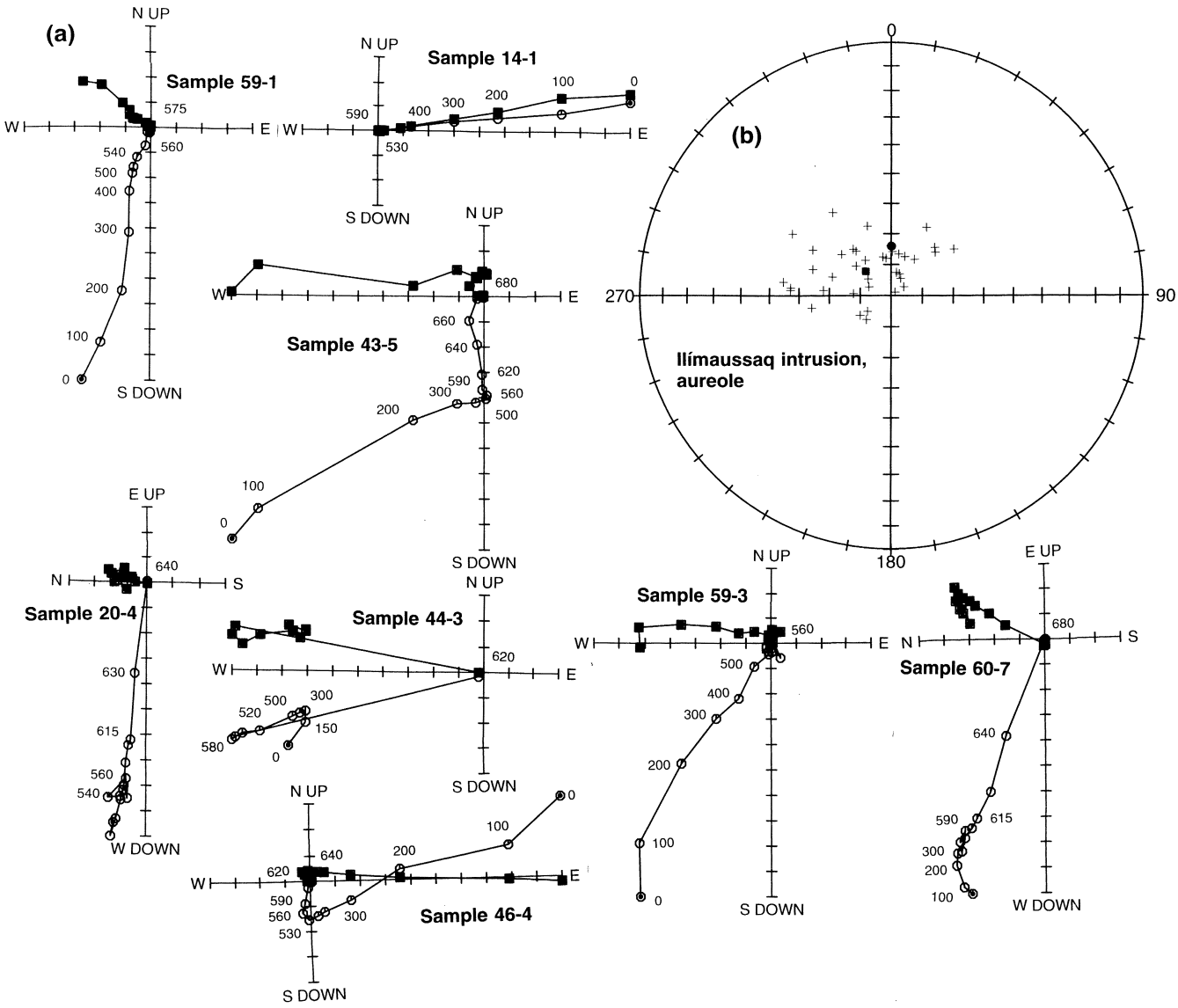


Figure 11. (a) Orthogonal projections illustrating thermal demagnetization of samples of Eriksfjord Group lavas (Ilímaussaq Formation) within the thermal aureole of the Ilímaussaq igneous complex; site numbers are after Thomas & Piper (1992) and symbols are as for Fig. 3. (b) Overprinted directions of magnetization in the aureole. The closed square is the group mean direction and the circle is the present-day field direction.

Table 3. Palaeomagnetic Results from the Ilímaussaq intrusion and contact aureole, Gardar Igneous Province, South Greenland.

Site code:	*N*	*D*	*I*	*R*	*k*	α_{95}
(i) *Syenites, northern sector*:						
IL1	5	291.5	80.5	4.84	25.5	15.5
IL2	5	358.4	68.2	4.97	146.2	6.3
IL3	5	353.9	71.3	4.97	156.7	6.1
IL4	5	1.1	74.5	4.98	178.4	5.7
IL5	8	5.3	77.2	7.94	123.5	5.0
All ChRMs	28	353.5	75.6	27.50	55.6	3.7
(Palaeomagnetic pole: 258.5°E, 86.4°N, $dp/dm = 6.2/6.8°$)						
(ii) *Ilímaussaq overprints in lavas of aureole*						
All ChRMs	38	312.8	78.2	36.28	21.5	5.1
(Palaeomagnetic pole: 262.1°E, 69.0°N, $dp/dm = 9.1/9.7°$)						
(iii) *Marginal syenites, southern sector*						
1	7	316.0	81.5	6.86	44.1	9.2
2	2	276.7	76.2	1.97	31.9	–
3	5	166.4	89.1	4.85	26.0	15.3
4	2	238.3	71.4	1.95	21.8	–
5	3	229.5	80.7	2.95	42.7	19.1
6	8	265.6	80.2	7.82	38.4	9.1
All ChRMs	27	269.4	82.6	26.23	34.0	4.8
(Palaeomagnetic pole: 286.2°E, 57.5°N, $dp/dm = 9.2/9.5°$)						
(iv) *Ilímaussaq Intrusion, Group Mean Results*:						
All sites	10	327.1	81.4	9.84	57.8	6.4
(Palaeomagnetic pole: 282.3°E, 72.6°N, $dp/dm = 12.0/12.4°$)						
All samples*	55	328.6	81.4	53.26	31.0	3.5
(Palaeomagnetic pole: 283.3°E, 72.9°N, $dp/dm = 6.6/6.8°$)						

Symbols are as for Table 1.
*Excluding shallow inclination components; see text.

contrasting observations have prompted us to reinvestigate the southern part of the complex. Since magnetizations were originally recognized by end-point analysis, which is not acceptable by modern standards, the new study was conducted on core end material from the six syenite and kakortakite sites of the original field sample but this time employing stepwise thermal demagnetization and component analysis.

The results (Fig. 12) confirm the dominance of magnetizations with a steep inclination. Because the ChRMs are steep, the declinations are variable but they range mostly from north to west to south (samples 4-3, 6-1 and 6-4, respectively, in Fig. 12). Interpretation of this remanence as a primary cooling-related magnetization is supported by recovery of inversely magnetized examples at sites 5 and 6 in the marginal syenites (sample 6-3 in Fig. 12). A few examples of both polarities have multi-component structures, with the steep ChRM removed by $\sim 500\,°C$ isolating a shallower W+(samples 5-1 and 6-1 in Fig. 12) or E–(sample 6–3) component in the higher part of the magnetite spectrum. These components have inclinations ranging from 35° to 60° but form no coherent population; they may therefore record sporadic later overprinting, possibly resulting from exsolution and oxidation of the magnetite. In Table 3, a group mean direction and pole position are calculated from the dual polarity steep positive components with $I > 65°$. The mean direction ($D/I = 269/83°$, $\alpha_{95} = 4.8°$, Table 3) has a similar westerly declination but slightly steeper inclination than the earlier result obtained by a.f. demagnetization and end-point analysis. The angle between the 22 normal ($D/I = 274/83°$, $\alpha_{95} = 5.6°$) and five reversed polarity components ($D/I = 87/-79°$, $\alpha_{95} = 11.2°$) is 4.3°; this is less than the critical angle of 11.5° and yields a positive class C reversal test (McFadden & McElhinny 1990).

There appears to be no tectonic explanation for the contrast in magnetic declination between the northern and southern sectors of the Ilímaussaq pluton: the complex was stoped into the metamorphic basement after the bulk of the Gardar rifting was complete. The quasi-horizontal layering of most of the interior agpaitic facies further supports an absence of significant tilting. Hence group mean directions and pole positions are listed in Table 3 by combining *in situ* sample and site-mean results from both sides of the pluton. We qualify this result by noting that data include a range of declinations that are probably a record of rapid palaeofield migration during the interval of primary cooling; this interpretation is supported by the observation that the range of declinations from the complex is embraced by the overprinted magnetizations in the thermal aureole (Fig. 11).

7 INTERPRETATION

7.1 Size and age of the Gardar–Keweenawan Loop

Palaeomagnetic data from the Gardar Province in South Greenland are summarized in Table 4. The case for preservation of a Late Mesoproterozoic–Early Neoproterozoic record in the bulk of these rocks is based on (1) the systematic temporal migration of the palaeofield defined by radiometrically dated igneous bodies assigned to the three (early, middle and late) stages of Gardar magmatism (Piper 1992); (2) the absence of a blanket remagnetization by the last phase of Gardar magmatism including the Ilímaussaq complex; (3) the demonstration of contact remagnetization adjacent to plutons (Piper 1992, 1994) and dykes (Thomas 1992); and (4) the absence of overprinting by the only significant later thermal event to influence this region, namely emplacement of dykes during Mesozoic rifting (Piper 1995). The collective data define an initial westward APW motion between *ca.* 1300 and 1260 Ma followed by reversal of this trend at 1260–1220 Ma, after which the APW executed a long northeasterly trending path referred to as the Gardar Track (Piper 1992).

Palaeopoles from Greenland can be compared with poles from the remainder of the Laurentian Shield in North America following adjustment for effects of late Cretaceous–early Tertiary seafloor spreading between Greenland and Labrador. Rotated poles are listed in Table 4 following adjustment to the reconstruction of Bullard *et al.* (1965, clockwise rotation of 18° about a Eulerian pole at 265.6°E, 70.5°N).

Most of the *ca.* 1300–1220 Ma interval recorded by earlier phases of Gardar magmatism was dominated by one field polarity characterized by westerly positive directions. The exception is the coherent sequence of multiple field inversions observed in the Eriksfjord lava succession (Piper 1977a; Thomas 1992; Thomas & Piper 1992). The repeated asymmetry of normal and reversed field directions observed here is indicative of a geomagnetic field source more complex than dipolar during the time interval embraced by the volcanic episode. Modelling of the field source from results at a single locality cannot be unique (Nevanlinna & Pesonen 1983; Thomas 1992), and pole positions calculated from the geocentric axial dipole assumption (as listed in Table 4 and plotted in Fig. 13) must therefore be qualified with this reservation.

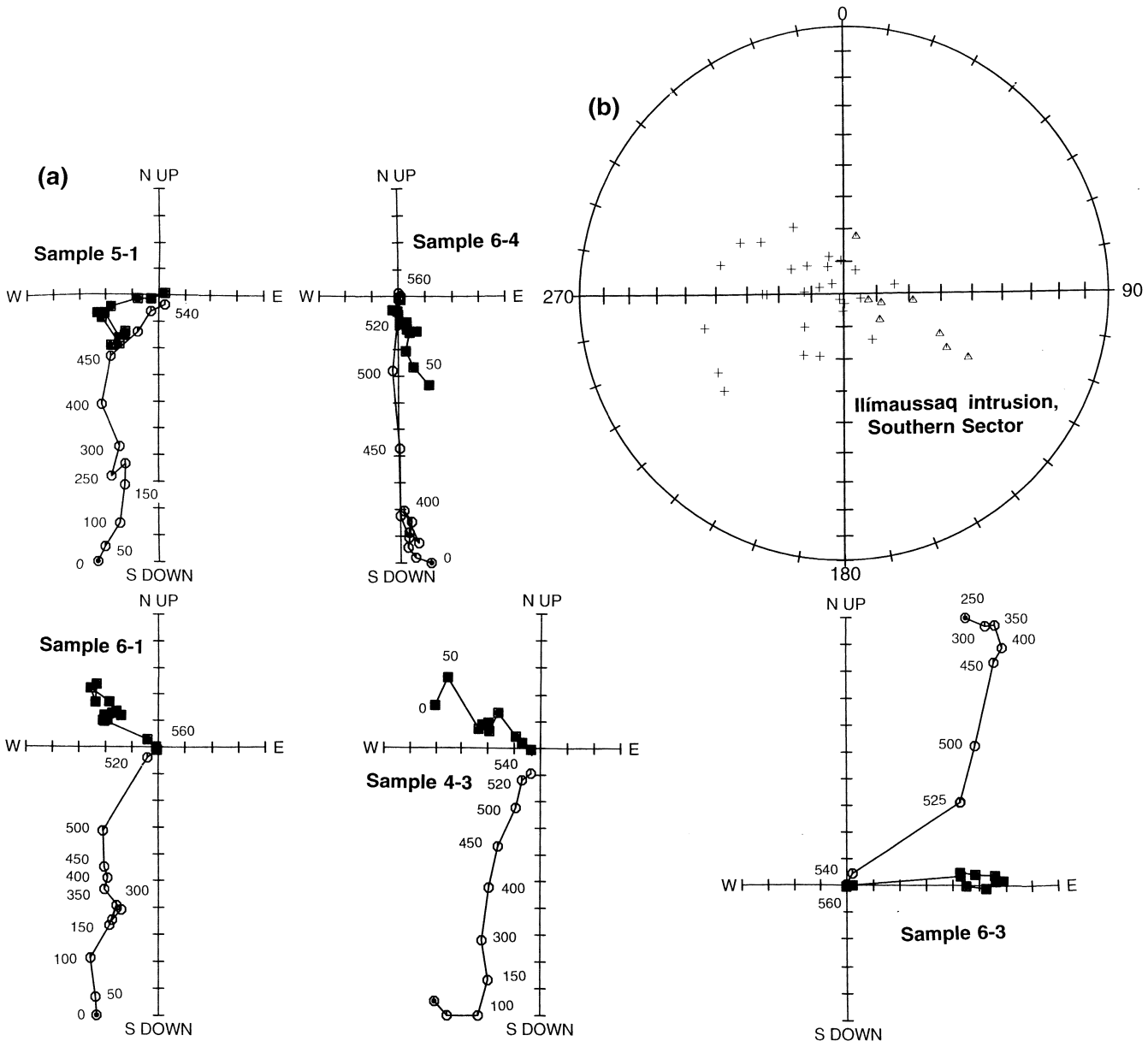


Figure 12. (a) Orthogonal projections illustrating thermal demagnetization behaviours of syenite samples from the southern sector of the Ilímaussaq intrusion at Tunugdliarfik (see arrowed location in Fig. 2b). The site numbers are after Piper (1976) and the symbols are as for Fig. 3. (b) ChRM directions from syenite samples in this region.

Nevertheless, these poles immediately succeed poles from North America dated *ca.* 1350 Ma (Piper 1987) on the APW path and are compatible with geological evidence (Upton & Emeleus 1987) for rift infilling shortly before emplacement of the earliest alkaline bodies in the Gardar Province at 1350–1290 Ma (Blaxland *et al.* 1978; Piper 1992; Cassi *et al.* 1993).

Whilst this segment of APW corresponds to results from the Grand Canyon Supergroup, the more significant correlation of 1290–1220 Ma results from the Gardar Province is with results from the Mackenzie Igneous Province (MIP). The MIP is currently dated at 1267 ± 2 Ma (LeCheminant & Heaman 1989) and incorporates dyke swarms spanning the width of the Laurentian Shield from the Sudbury dyke swarm at the SE margin to the Muskox and Coppermine igneous events in the NW. Dyking in South Greenland embraced by the mid-Gardar episode is probably a component of this shield-wide rifting event (Macdonald & Upton 1993). Gardar rocks are dominated by the same geomagnetic field polarity as that recorded by the MIP, and the MIP poles plot marginally to the SW of poles from the mid-Gardar dykes (Piper 1995). In the context of Rb–Sr ages of 1254–1203 Ma from the BD0 and BD1 members of Gardar dyke swarms (Patchett *et al.* 1978), the radiometric evidence suggests that these poles could be between 25 and 70 Ma younger than the MIP poles. The collective evidence is therefore indicative of only small APW motion for at least 50 Ma between ~1270 and 1220 Ma.

This quasi-static interval was succeeded at some time after 1220 Ma by rapid APW movement, so that by the time of the mid-Gardar igneous episode at *ca.* 1160 Ma the pole

Table 4. Summary of palaeomagnetic poles from the Gardar Province of South Greenland before and after rotation to North America.

Pole Rock Unit	Age (Ma)	D/I	Pole Position °E	°N	dp/dm	Si(sa)	Rotated Pole °E	°N	Ref.
1. Eriksfjord Group Volcanics:									
(a) Basal Lavas and carbonatite		290/1	207	10	7/14	5(36)	191	16	1
(b) Lower Lava Group*		260/25	239	7	7/14	7(34)	223	10	2,3
(c) Upper Lava Group*		280/24	222	16	5/9	24(113)	206	21	2,3
2. Grønnedal-Íka Complex	1299 ± 17	271/21	228	9	3/5	10(53)	212	13	4
3. Motzfeldt Complex	1350 ± 10	286/4	212	10	6/12	2(25)	196	15	1
4. North Qôroq Complex	1291 ± 61	297/2	201	14	3/6	12(65)	185	20	1
5. Narssassuaq Stock		288/1	207	9	4/7	3(23)	191	15	1
6. Lamprophyre Dykes		285/−9	206	3	5/10	9(51)	189	9	5
7. BDO Dyke		84/+19	203	−12	3/5	1(8)	186–6	6	
8. Kungnât Complex	1219 ± 17	292/−15	199	3	2/4	4(18)	182	9	5
BD1 Dykes, western sector		284/−7	308	4	8/16	10(48)	291	1	6
9. BD1 Dykes, north-west sector		277/−1	215	3	5/10	21(126)	198	8	6
10. Dolerite dykes in Grønnedal Complex		275/12	220	7	4/9	5(32)	204	12	6
BD2-BD3 Dykes, west sector		287/43	220	30	5/7	17(86)	205	35	6
11. BD2-BD3 Dykes, NW sector		288/56	227	40	9/13	13(71)	214	44	6
Dyke overprint in lavas, Narssassuaq		267/48	229	25	6/9	6(35)	214	29	1
12. Hviddal (Older) Giant Dyke, Tugtutôq	1154 ± 9	294/43	215	33	8/12	7(38)	200	38	7
13. Narssaq Gabbro and Ultramfics		285/47	225	32	8/12	4(22)	211	36	7
14. Younger Gabbro Giant Dykes, Tugtutôq		292/57	226	42	8/11	13(73)	213	46	7
15. South Qôroq Complex	1160 ± 8	301/51	214	42	5/7	9(67)	200	47	1
Grønnedal Complex, overprint		303/51	210	43	9/13	(13)	196	48	4
Motzfeldt Complex, overprint		311/51	205	47	8/13	(8)	191	53	1
16. BD3 Dyke Swarms, Eastern Province (a)		290/51	224	36	4/6	47(246)	210	40	5
(b)		283/51	231	33	5/7	18(102)	217	37	6
Amphibolite overprint, Grønnedal region		283/60	235	41	7/9	(22)	222	45	4
Eriksfjord Group Red beds B		284/67	244	48	8/9	(31)	233	51	8
17. Steep I dykes, western sector		262/75	268	48	22/24	4(25)	258	48	6,8
18. Ilímaussaq Complex 1130 ± 50									
(a) southern part		269/83	286	58	9/10	(27)	278	56	8
(b) northern aureole		313/78	262	69	8/10	(38)	261	69	8
(c) northern part		354/76	259	86	6/7	(28)	309	84	8
(d) pluton, mean		327/81	282	73	12/12	10(55)	283	71	8
Eriksfjord Group Red beds A		305/34	202	32	3/5	(57)	186	38	8
Eriksfjord Group Red beds C		179/38	315	−8	6/10	(17)	297–11		8

References to palaeomagnetic studies are: 1, Piper (1992); 2, Piper (1977a); 3, Thomas & Piper (1992); 4, Piper (1994); 5, Piper & Stearn (1977); 6, Piper 1995; 7, Piper (1977b); 8, This paper. Si(sa): number of sites (samples). Studies published in the 1970s are based on directions resolved by progressive a.f. cleaning; most identify only one polarity. Quality factors (Q) are therefore mostly 3 or 4 (Van der Voo 1993). Studies published in the 1990s are based on progressive thermal cleaning, or a mixture of thermal and a.f. cleaning, accompanied by component analysis; some include contact tests. Rotated poles are calculated following closure of the Davis Strait by a combined rotation of Greenland to Europe and Europe/Greenland to North America amounting to 18° clockwise rotation about a Euler Pole at 265.6°E, 70.5°N according to the Bullard *et al.* (1965) reconstruction. * Repeated asymmetry of normal and reversed field axes recorded in these lavas implies a more complex geomagnetic source than the axial geocentric dipole model used to compute the poles.

had moved 50° to the northeast (poles 13–18 in Fig. 13). Inclusion of the Ilímaussaq pole doubles this motion between ~1160 and ~1130 Ma (i.e. during the last phase of Gardar magmatism) to execute the older limb of a flattened APW loop incorporating the Gardar and Keweenawan Tracks. The polar shift between 1220 Ma and 1160 Ma is recorded by dyke emplacement in SW Greenland as a progressive increase in inclination and small clockwise rotation of the declination in the magnetizations of successive BD1, BD2 and BD3 dyke swarms (Piper 1995 and Table 4); the dated major plutons in both eastern and western parts of the province define a coherent sequence of poles along the same track (poles 3, 4, 8, 12, 15 and 18 in Fig. 13 and Table 4).

There appears to be no record of this ~90° long Gardar Track in the record from North America. The ensuing path after ~1100 Ma is, however, well defined from the Keweenawan Province. The essential features of the Keweenawan Track are shown in Fig. 13 and include an inversion from predominant reversed polarity to exclusive normal polarity as the path moves southwestwards across the northwest Pacific (Fig. 13). Ongoing motion carries the APW path back to a similar position to that occupied at 1270–1220 Ma by the interval 1100–1040 Ma (Palmer & Davies 1987). The continuation of the track is defined by the sequence of uplift and cooling magnetizations within the Grenville tectonothermal Province (see Fig. 1 and summary of poles in Fig. 13).

The oldest pole positions defining the northeastern extremity of the Keweenawan Track appear to come from rift-related alkaline igneous bodies emplaced within the Kapuskasing Structural Zone cutting the Superior Province (Symons 1989, 1992); they are dated between 1109 and 1098 Ma and plot ~30° to the southwest of the Ilímaussaq pole. We therefore

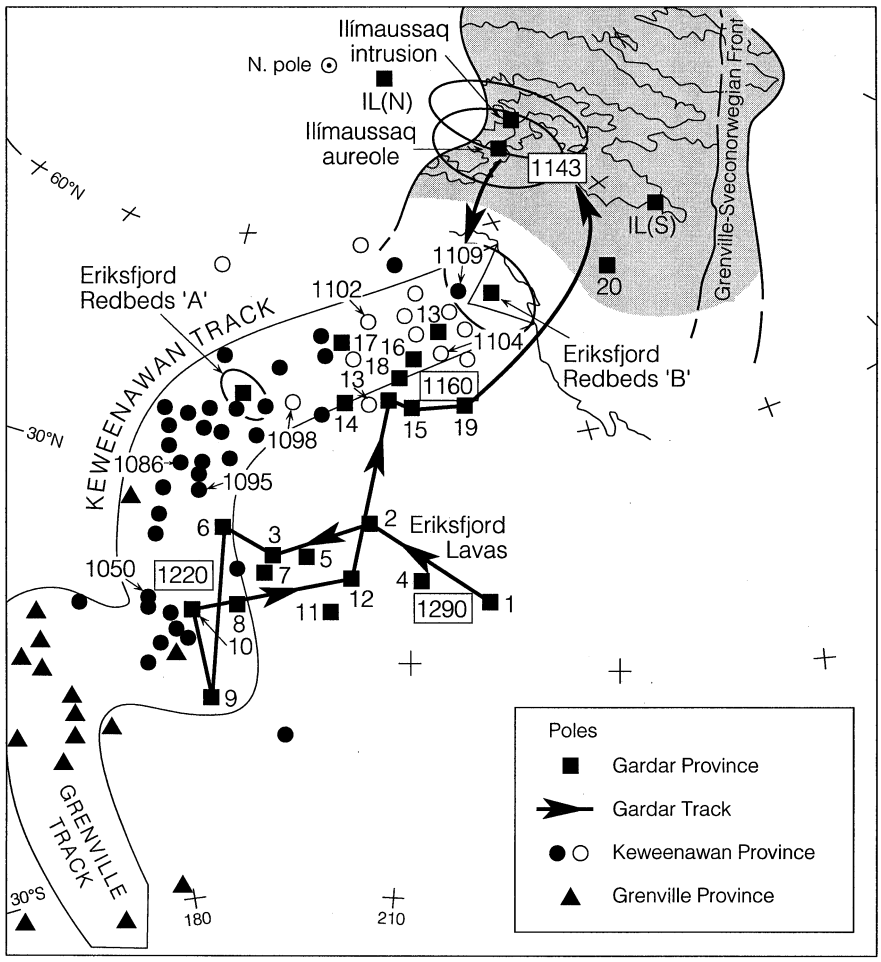


Figure 13. Palaeomagnetic poles derived from sequential geological episodes in the Gardar Province of South Greenland. A complete summary of the palaeomagnetic poles from this province is given in Table 4. The poles interpreted to be derived from primary magnetizations are plotted here in time sequence (black squares) and are numbered: 1, Eriksfjord Group, lower lavas (~1350–1300 Ma); 2, Eriksfjord Group, upper lavas; 3, Narssassuaq Stock; 4, Grønnedal-Íka Complex (1299 ± 17 Ma); 5, Motzfeldt Complex (1291 ± 31 Ma); 6, North Qôroq Complex (1291 ± 61 Ma); 7, basal lavas and carbonatite, Eriksjord Group (slightly older than poles 1 and 2); 8, Lamprophyre dykes; 9, BDO dyke; 10, Kungnât Complex (1219 ± 17 Ma); 11, BD1 dykes; 12, dolerite dykes in Grønnedal Complex; 13, DB2-BD3 dyke swarm (two poles); 14 Hviddal (older) giant dyke, Tugtutôq (1154 ± 9 Ma); 15, Narssaq gabbro and ultramafics; 16, younger gabbro giant dykes, Tugtutôq; 17, South Qôroq Complex (1160 ± 8 Ma); 18 and 19, BD3 dyke swarms; 20, DB3 dykes, steep inclination group. The poles are rotated into a North American reference frame to adjust for Phanerozoic drift between North America and Greenland. North American poles from the Keweenawan Province of reversed polarity (open circles) and normal polarity (closed circles), and from the Grenville Province (triangles) are shown for comparison.

conclude that the rapid north-easterly motion of the pole embraced by the Gardar Track terminated after ~1130 Ma. Subsequently, and possibly after a near-static period, it reversed to follow the southwesterly Keweenawan Track. The rotation of palaeomagnetic declination in the Ilímaussaq intrusion may record 'hairpin' motion near the junction of the Gardar and Keweenawan tracks; the degree of differentiation in this complex is indicative of protracted cooling and possibly magnetization. Much more work, however, will be required on rocks dated 1140–1110 Ma to define properly the join between the two tracks.

Outward and return paths of the Gardar–Keweenawan Loop are close together but, with the exception of convergences at ~1160 and ~1250 Ma, the return path is displaced to the northwest. It is therefore potentially possible to date unconstrained magnetizations by noting their correspondence to one or other of these tracks.

7.2 Magnetization of the Eriksfjord Group red beds

Currently, the only direct age dates from the Eriksfjord succession are two Sm–Nd determinations of 1200 and 1170 Ma on lavas from the Ulukasik Formation (Cassi *et al.* 1993). We consider that these determinations are unlikely to represent emplacement ages for four reasons.

(1) Outliers of the Eriksfjord Group in the northeast of the outcrop (Fig. 1) are cut by the Motzfeldt (1350–1310 Ma) and North Qôroq (1291 Ma) intrusions. Lavas in an outlier cut by the Motzfeldt body correspond very closely to lavas of the Ilímaussaq Formation (Jones 1980).

(2) There is no evidence for a significant disconformity within the Eriksfjord Group, as would be required if the basal members were significantly older than the dated Ulukasik Formation. The rift infill is dominated by coarse clastics; lava

eruption, at least in the Ulukasik Formation (Thomas & Piper 1994), was very rapid, and the entire magmatic system feeding the rift infill is unlikely to have lasted more than a few millions of years (Section 2).

(3) The pole positions from the Eriksfjord lavas are compatible with other poles from the Laurentian Shield in the interval 1350–1220 Ma (Piper 1987). By ~1170 the APW path had moved 30–40° to the northeast (Fig. 13) and shows no correspondence with the Eriksfjord poles (*cf*. Cassi *et al*. 1993).

(4) The Eriksfjord lava succession exhibits multiple reversals of the ancient field (Piper 1977a; Thomas & Piper 1992). However, Laurentian results assigned to the interval 1290–1220 Ma and again during the mid-Gardar event at ~1160 Ma show exclusive normal polarity (Piper 1995). Mixed polarities are, however, recognized during the interval 1400–1300 Ma.

In view of the above evidence, we suggest than the Sm–Nd determinations from the Ulukasik Formation record resetting during the voluminous mid-Gardar magmatic episode (see also Cassi *et al*. 1993). The geological evidence (Section 2) supports the view that the Eriksfjord Group was deposited in the early stages of the Gardar tectonomagmatic episode at ~1350–1300 Ma. The magnetizations derived from the red sediments show no correspondence with the APW path during these times (Fig. 13).

Although magnetizations in the red beds can only otherwise be recognized as post-lithification events from the field evidence, the APW path described in Section 7.1 forms a basis for dating these magnetizations. The Eriksfjord 'A' magnetization has a declination rotated too far in a clockwise sense to correlate with other directions of magnetization from the Gardar Province (Table 4). It does, however, yield a pole position plotting on the Keweenawan Track together with North American poles dated a little younger than 1100 Ma in age. This is close to the age of the Grenville metamorphism (Berger & York 1981) and it is logical to seek a causitive link between the magnetization and the Grenville orogen formerly sited 100–200 km to the south of this study area (Fig. 1).

The 'B' magnetization has a mean inclination which is slightly steeper than the mid-Gardar igneous episode (including the NE–SW dyking following the axial trend of the rift) but shallower than remanence in the Ilímaussaq complex. Clearly, emplacement of the Ilímaussaq intrusion has not played a major part in overprinting the sediments although steeper inclination magnetizations included within the 'B' population could have resulted in this way (*cf*. Fig. 6 and Figs 11 and 12). From the position of the 'B' group mean pole on the Gardar–Keweenawan Loop, it is concluded that this general remanence may have been acquired:

(1) during waning stages of the mid-Gardar igneous event (which included dyke swarms, emplacement of giant dykes and gabbro/ultrabasic plutons, see Table 4), but mostly before emplacement and cooling of the Ilímaussaq complex, or

(2) some 20 Ma after all igneous activity had ceased in the Gardar, although contemporaneous with some of the Keweenawan activity (Fig. 13).

Three possible origins for both overprints 'A' and 'B'. are (1) a thermoviscous magnetization (TVRM) acquired during burial and subsequent uplift (2) a thermochemical remanence (TCRM) acquired by magnetite and haematite precipitated by fluid circulation motivated by igneous heating, or (3) a purely chemical magnetization (CRM) acquired by precipitation of haematite during diagenesis.

There are several indirect indications for an overlying cover to the present outcrop: lavas presently exposed at the top of the Eriksfjord Group are metamorphosed to lower greenschist facies, and dyke densities remain high up to this level. Furthermore, differentiation in the Ilímaussaq intrusion occurred within a closed system implying the presence of a roof zone of some mechanical strength (Professor B. G. J. Upton, personal communication). These considerations suggest a possible cover of perhaps 2 km in Gardar times. The maximum temperatures attained by any plausible geothermal gradient at a depth of ~5000 m near the base of the sedimentary pile would, however, have been well below the unblocking temperatures recognized by demagnetization; the theoretical curves of Pullaiah *et al*. (1975) and Middleton & Schmidt (1982) indicate that temperatures operating over geological time are incapable of producing a TVRM in this situation. Hence the acquisition of TVRM can probably be discounted.

The second mechanism appears to be a plausible explanation for the 'B' magnetization: this is confined to the base of the succession, and the polarity and mean palaeomagnetic direction are compatible with an age only marginally younger than magmatism, including a giant dyke complex and extensive minor dyking, within the rift at ~1160 Ma (Fig. 13). The thermal anomaly within the rift would have motivated fluid flow through lower levels of the sedimentary pile with upward passage effectively sealed by the overlying lavas; the chemistry of such a deep and sealed system is likely to have favoured the precipitation of magnetite, which is the main carrier of the 'B' remanence. This explanation is also consistent with the probable relative importance of the igneous events influencing the Eriksfjord Group. The gabbro giant dyke complex has a gross thickness of ~1 km and an overall length of at least 140 km. On the assumption that this extends to the base of the crust, more than 3000 km^2 of basaltic magma was emplaced close to the rift axis in mid-Gardar times (Professor B. G. J. Upton, personal communication). In contrast, the Ilímaussaq intrusion, which has had no obvious impact on the magnetization of the Eriksfjord sediments, is of relatively insignificant volume representing merely the topmost slice in the culmination of a deeper-lying basic body (see Stevenson *et al*. 1997).

The 'A' magnetization, which is dominant in the higher beds, is most likely to be a pure CRM: it resides in haematite and has a direction contrasting with adjoining igneous rocks. Furthermore, the proximity of the pole to the time of Grenville orogenesis suggests a link with this event. Fluids containing abundant hydrated minerals are expelled from orogens into the continental forelands following burial beneath overthrust nappes (e.g. Miller & Kent 1988), and large volumes of fluid, perhaps as much as half of the volume of the overthrust units, can be involved in this migration and react with the host lithologies. The precipitated minerals are controlled by ambient pH–Eh conditions, and reactions include the release of iron into solution which can then contribute to the precipitation of ferromagnetic minerals (Oliver 1986). However, the examples where this mechanism can plausibly be invoked were sealed environments in which a mildly alkaline and reducing chemistry resulted in the precipitation of magnetite. In contrast, diagenesis producing haematite occurs in oxidizing environments and is promoted in systems with access to the surface.

In the Irish Sea, for example, it post-dates Variscan orogenesis and correlates with extensional tectonism (Piper & Crowley 1999). In the Gardar Province a link between the 'A' remanence and the Grenville orogenesis is therefore expected to correlate with the extensional collapse rather than orogenic loading. The source of the haematite could then be due to breakdown of ferromagnesian silicates or fixing of a precursor ferromagnetic mineral during local fluid migration. It occurred predominantly within porous sandstones of the Naujurssuit Formation and would not have necessitated large-scale lateral fluid migration.

It remains to note possible interpretations of the 'C' magnetization observed in the Eriksfjord Group. This correlates with no part of the Gardar, Keweenawan or Grenville records and is therefore inferred to be younger in age. It is difficult to isolate an obvious geological cause because this region was in a quiescent cratonic setting during Phanerozoic times. Although the influence of Late Precambrian–Early Cambrian rifting is recognized in West Greenland (Piper 1981a,b), the pole from the 'C' magnetization is removed from North American poles of this age (Meert *et al*. 1994). This remanence also contrasts with the magnetizations in Mesozoic dyke swarms emplaced into the continental margin of South Greenland (Piper 1995). The rotated pole (297°E, 11°S) does, however, plot with Silurian and Lower Carboniferous poles from North America (Van der Voo 1993) and therefore seems most likely to record a weak overprinting effect of the Acadian Episode in Palaeozoic times. As with the 'A' magnetization, an origin by chemical authigenesis is the most likely explanation; a link to the Acadian orogeny is suggested by analogy with late Palaeozoic overprints north of the orogenic front in Newfoundland (Deutsch & Prasad 1987).

ACKNOWLEDGMENTS

Fieldwork for this study was supported by the Natural Environment Research Council, and Mr Ole Plesner kindly provided much logistical help in South Greenland. We are grateful to Dr Jim Parsons for initiating the palaeomagnetic study and to Dr Tim Rolph for his valuable advice during the laboratory investigations. Kay Lancaster is thanked for drafting several of the figures and we are grateful to Professor B. G. J. Upton and Dr H. C. Palmer for their critical reviews of the manuscript. ZQR's visit to the Geomagnetism Laboratory was supported by the Royal Society.

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